

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).
- Remove the lubricating oil filters. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>).
- Remove the fuel filters, if needed. Refer to Procedure 006-015 (</qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html>).
- Remove the fuel filter head. Refer to Procedure 006-017 (</qs3/pubsys2/xml/en/procedures/56/56-006-017-tr.html>).
- Disconnect the pre-lubrication suction hose. Refer to Procedure 007-063 (</qs3/pubsys2/xml/en/procedures/56/56-007-063-tr.html>).

- Remove the oil pan from the oil pan adapter. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>).
- Remove the cover plates from the oil pan adapter, if needed. Refer to Procedure 007-026 (</qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html>).

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- Disconnect the pre-lubrication suction hose. Refer to Procedure 007-063

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- Remove the oil pan from the oil pan adapter. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html).
- Remove the cover plates from the oil pan adapter, if needed. Refer to Procedure 007-026 (/qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html).
- Disconnect the crankcase breather hoses from the oil pan adapter. Refer to Procedure 003-001 (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html).
- Remove the lubricating oil dipstick. Refer to Procedure 007-011 (/qs3/pubsys2/xml/en/procedures/56/56-007-011-tr.html).

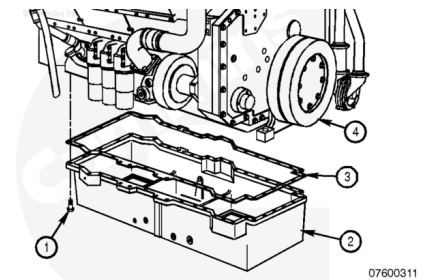
Remove

The clearance **must** be sufficient from the oil suction tube before the adapter can be removed from the bottom of the engine.

The QSK45 oil pan adapter assembly is held on by 36 capscrews, and the QSK60 oil pan adapter assembly is held on by 46 capscrews.

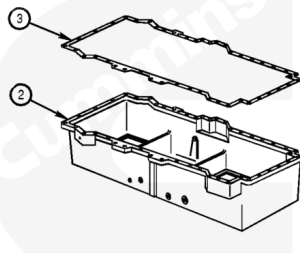
Remove the capscrews (1) from the oil pan adapter assembly (2), the gasket (3), and the engine (4).

Remove the oil pan adapter assembly (2) and the gasket (3).



Clean and Inspect for Reuse

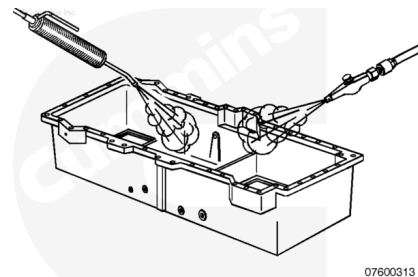
Inspect the gasket (3) for reuse.



07600312

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



07600313

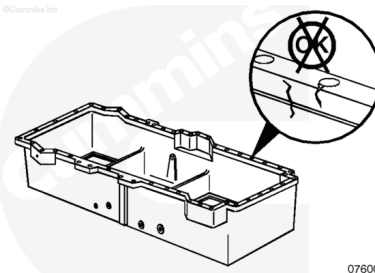
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean all gasket surfaces on cover plates, sumps, and oil pan adapter.

Steam clean the oil pan adapter and dry with compressed air.

Inspect all areas of the oil pan adapter for cracks or other damage. If any area is cracked or damaged, the oil pan adapter **must** be replaced.



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Install

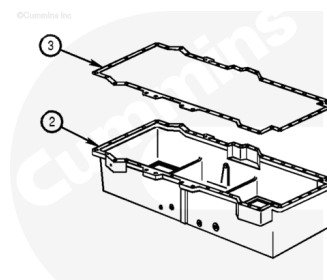
QSK45 Engines

⚠ CAUTION ⚠

Do not use gasket cement. Gasket cement will prevent the gasket from sealing properly.

To ease in assembly, plastic wire ties can be used to retain the gasket until the capscrews can be installed, then cut and remove the ties before tightening the capscrews.

Install the gasket (3) on the oil pan adapter assembly (2).

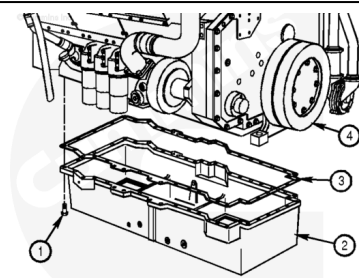


07600312

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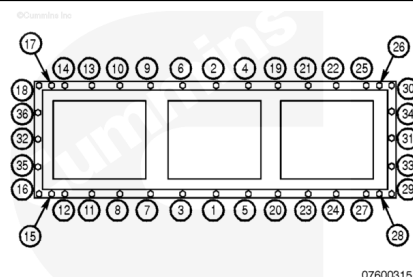
Position the oil pan adapter assembly (2) with the gasket (3) on the engine (4) with the capscrews (1).



07600311

Tighten the capscrews in the sequence shown.

Torque Value: 45 n·m [33 ft·lb]



07600315

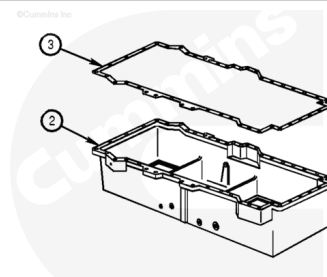
QSK60 Engines

⚠ CAUTION ⚠

Do not use gasket cement. Gasket cement will prevent the gasket from sealing properly.

To ease in assembly, plastic wire ties can be used to retain the gasket until the capscrews can be installed, then cut and remove the ties before tightening the capscrews.

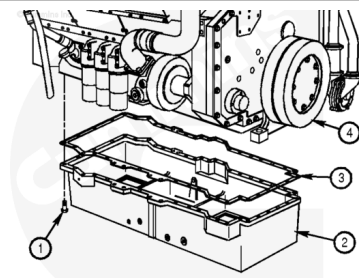
Install the gasket (3) on the oil pan adapter assembly (2).



07600312

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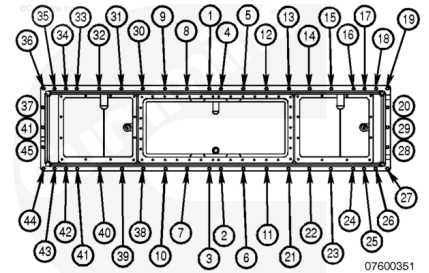


07600311

Position the oil pan adapter assembly (2) with the gasket (3) on the engine (4) with the capscrews (1).

Tighten the capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

with Mechanically Actuated Injector

- Install the oil pan onto the oil pan adapter. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>).
- Install the cover plates on the oil pan adapter, if removed. Refer to Procedure 007-026 (</qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html>).
- Install the fuel filter head. Refer to Procedure 006-017 (</qs3/pubsys2/xml/en/procedures/56/56-006-017-tr.html>).
- Install the new fuel filters, if removed. Refer to Procedure 006-015 (</qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html>).
- Install the new lubricating oil filters. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>).
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).
- Connect the pre-lubrication suction hose. Refer to Procedure 007-063 (</qs3/pubsys2/xml/en/procedures/56/56-007-063-tr.html>).
- Operate the engine at idle speed and inspect for oil leaks.



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- Install the fuel filter head. Refer to Procedure 006-017 (</qs3/pubsys2/xml/en/procedures/56/56-006-017-tr.html>).
- Install the new fuel filters, if removed. Refer to Procedure 006-015 (</qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html>).
- Install the new lubricating oil filters. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>).
- Install the pre-lubrication suction hose. Refer to Procedure 007-063 (</qs3/pubsys2/xml/en/procedures/56/56-007-063-tr.html>).
- Install the crankcase breather hoses to the oil pan adapter. Refer to Procedure 003-001 (</qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html>).
- Install the lubricating oil dipstick. Refer to Procedure 007-011 (</qs3/pubsys2/xml/en/procedures/56/56-007-011-tr.html>).
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).
- Operate the engine at idle speed and inspect for oil leaks.

